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6 Interconnect

6.1 Introduction

The wires linking transistors together are called *interconnect* and play a major role in the performance of modern systems. In the early days of VLSI, transistors were relatively slow. Wires were wide and thick and thus had low resistance. Under those circumstances, wires could be treated as ideal equipotential nodes with lumped capacitance. In modern VLSI processes, transistors switch much faster. Meanwhile, wires have become narrower, driving up their resistance to the point, that in many signal paths, the wire RC delay exceeds gate delay. Moreover, the wires are packed very closely together and thus a large fraction of their capacitance is to their neighbors. When one wire switches, it tends to affect its neighbor through capacitive coupling; this effect is called *crosstalk*. Wires also account for a large portion of the switching energy of a chip. On-chip interconnect inductance had been negligible but is now becoming a factor for systems with fast edge rates and closely packed busses. Considering all of these factors, circuit design is now as much about engineering the wires as the transistors that sit underneath.

The remainder of this section defines the dimensions used to describe interconnect and gives a practical example of wire stacks in nanometer processes. Section 6.2 explores how to model the resistance, capacitance, and inductance of wires. Section 6.3 examines the impact of wires on delay, energy, and noise. Section 6.4 considers the tools at a designer's disposal for improving performance and controlling noise. Section 6.5 extends the method of Logical Effort to give insights about designing paths with interconnect.

6.1.1 Wire Geometry

Figure 6.1 shows a pair of adjacent wires. The wires have width w, length l, thickness t, and spacing of s from their neighbors and have a dielectric of height h between them and the conducting layer below. The sum of width and spacing is called the wire *pitch*. The thickness to width ratio t/w is called the *aspect ratio*.

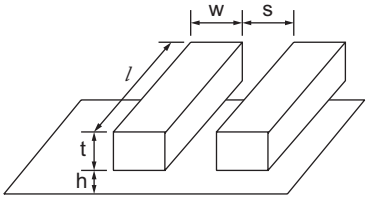


FIGURE 6.1 Interconnect geometry

Early CMOS processes had a single metal layer and until the early 1990s only two or three layers were available, but with advances in chemical-mechanical polishing it became far more practical to manufacture many metal layers. As discussed in Section 3.4.2, aluminum (Al) wires used in older processes gave way to copper (Cu) around the 180 or 130 nm node to reduce resistance. Soon after, manufacturers began replacing the SiO_2 insulator between wires with a succession of materials with lower dielectric constants (*low-k*) to reduce capacitance. A 65 nm process typically

has 8–10 metal layers and the layer count has been increasing at a rate of about one layer every process generation or two.

6.1.2 Example: Intel Metal Stacks

Figure 6.2 shows cross-sections of the metal stacks in the Intel 90 and 45 nm processes, shown to scale [Thompson02, Moon08]. The 90 nm process has six metal layers, while the 45 nm process shows the bottom eight metal layers. The transistors are tiny gizmos beneath the vast labyrinth of wire. Metal1 is on the tightest pitch, roughly that of a contacted transistor, to provide dense routing within cells. The upper levels are progressively thicker and on a greater pitch to offer lower-resistance interconnections over progressively longer distances. The wires have a maximum aspect ratio of about 1.8.

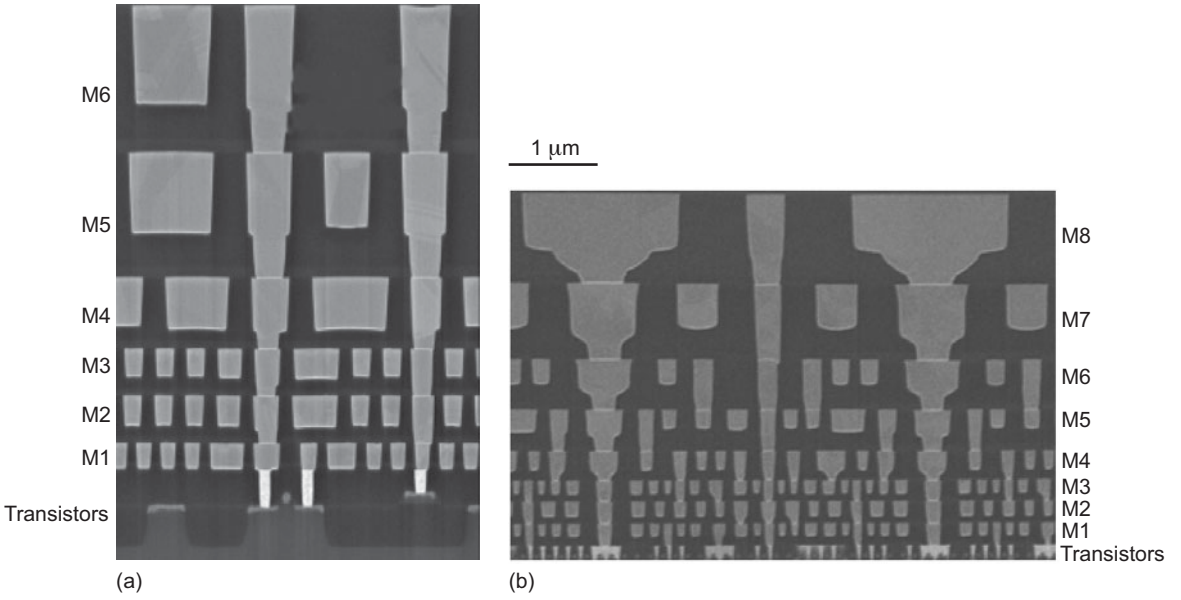


FIGURE 6.2 SEM image of wire cross-sections in Intel's (a) 90 nm and (b) 45 nm processes ((a) From [Thompson02] © 2002 IEEE. (b) From [Moon08] with permission of Intel Corporation.)

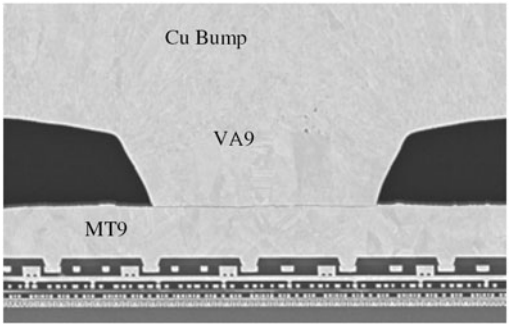


FIGURE 6.3 SEM image of complete cross-section of Intel's 45 nm process including M9 and I/O bump (From [Moon08] with permission of Intel Corporation.)

The top-level metal is usually used for power and clock distribution because it has the lowest resistance. Intel's 45 nm process introduced an unusual extra-thick ninth Cu metal layer used to distribute power to different power-gated domains across the die (see Section 5.3.2). Figure 6.3 shows a full cross-section including this MT9 layer, a Cu bump for connecting to the power or ground network in the package (see Section 13.2.2), and a VA9 via between MT9 and the bump. The lower levels of metal and transistors are scarcely visible beneath these fat top layers. Table 6.1 lists the thickness and minimum pitch for each metal layer.

TABLE 6.1 Intel 45 nm metal stack

Layer	*t* (nm)	*w* (nm)	*s* (nm)	pitch (nm)
M9	7 μm	17.5 μm	13 μm	30.5 μm
M8	720	400	410	810
M7	504	280	280	560
M6	324	180	180	360
M5	252	140	140	280
M4	216	120	120	240
M3	144	80	80	160
M2	144	80	80	160
M1	144	80	80	160

6.2 Interconnect Modeling

A pipe makes a good mechanical analogy for a wire, as shown in Figure 6.4 [Ho07]. The resistance relates to the wire's cross-sectional area. A narrow pipe impedes the flow of current. The capacitance relates to a trough underneath the leaky pipe that must fill up before current passes out the end of the pipe. And the inductance relates to a paddle wheel along the wire with inertia that opposes changes in the rate of flow. Each of these elements is discussed further in this section.

A wire is a distributed circuit with a resistance and capacitance per unit length. Its behavior can be approximated with a number of lumped elements. Three standard approximations are the L-model, π-model, and T-model, so-named because of their shapes. Figure 6.5 shows how a distributed RC circuit is equivalent to N distributed RC segments of proportionally smaller resistance and capacitance, and how these segments can be modeled with lumped elements. As the number of segments approaches infinity, the lumped approximation will converge with the true distributed circuit. The L-model is a poor choice because a large number of segments are required for accurate results. The π-model is much better; three segments are sufficient to give results accurate to 3% [Sakurai83]. The T-model is comparable to the π-model, but produces a circuit with one more node that is slower to solve by hand or with a circuit simulator. Therefore, it is common practice to model long wires with a 3–5 segment π-model for simulation. If inductance is considered, it is placed in series with each resistor. The remainder of this section considers how to compute the resistance, capacitance, and inductance.

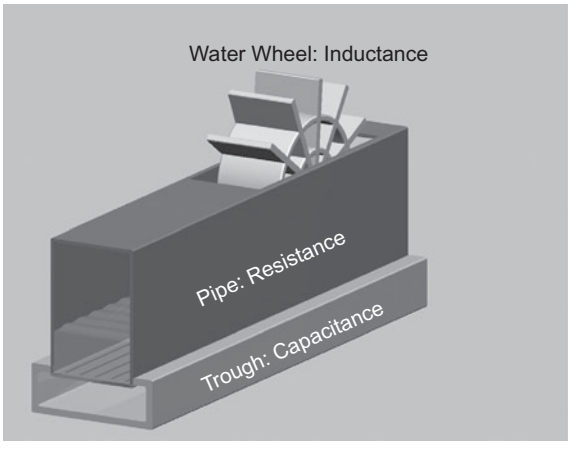


FIGURE 6.4 Pipe analogy for wire

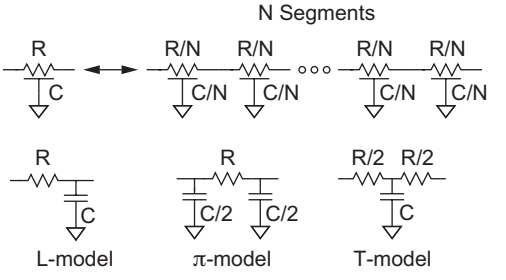


FIGURE 6.5 Lumped approximation to distributed RC circuit

6.2.1 Resistance

The resistance of a uniform slab of conducting material can be expressed as

$$R = \frac{\rho}{t}\frac{l}{w} \tag{6.1}$$

where ρ is the resistivity.[1] This expression can be rewritten as

$$R = R_{\square}\frac{l}{w} \tag{6.2}$$

where $R_{\square} = \rho/t$ is the *sheet resistance* and has units of Ω/square. Note that a square is a dimensionless quantity corresponding to a slab of equal length and width. This is convenient because resistivity and thickness are characteristics of the process outside the control of the circuit designer and can be abstracted away into the single sheet resistance parameter.

To obtain the resistance of a conductor on a layer, multiply the sheet resistance by the ratio of length to width of the conductor. For example, the resistances of the two shapes in Figure 6.6 are equal because the length-to-width ratio is the same even though the sizes are different. Nonrectangular shapes can be decomposed into simpler regions for which the resistance is calculated [Horowitz83].

Table 6.2 shows bulk electrical resistivities of pure metals at room temperature [Bakoglu90]. The resistivity of thin metal films used in wires tends to be higher because of scattering off the surfaces and grain boundaries, e.g., 2.2–2.6 $\mu\Omega \cdot$ cm for Cu and 3.6–4.0 $\mu\Omega \cdot$ cm for Al [Kapur02].

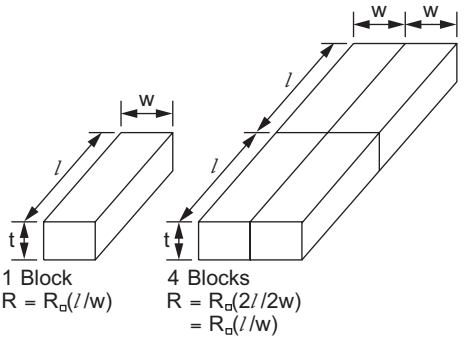


FIGURE 6.6
Two conductors with equal resistance

TABLE 6.2 Bulk resistivity of pure metals at 22 °C

Metal	Resistivity ($\mu\Omega \cdot$ cm)
Silver (Ag)	1.6
Copper (Cu)	1.7
Gold (Au)	2.2
Aluminum (Al)	2.8
Tungsten (W)	5.3
Molybdenum (Mo)	5.3
Titanium (Ti)	43.0

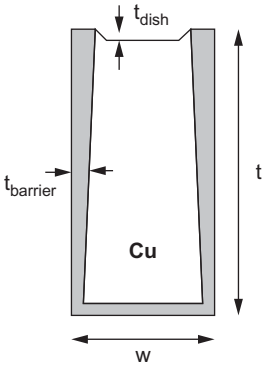


FIGURE 6.7 Copper barrier layer and dishing

As shown in Figure 6.7, copper must be surrounded by a lower-conductivity diffusion barrier that effectively reduces the wire cross-sectional area and hence raises the resistance. Moreover, the polishing step can cause *dishing* that thins the metal. Even a 10 nm barrier is quite significant when the wire width is only tens of nanometers. If the average barrier thickness is t_{barrier} and the height is reduced by t_{dish}, the resistance becomes

$$R = \frac{\rho}{\left(t - t_{\text{dish}} - t_{\text{barrier}}\right)}\frac{l}{\left(w - 2t_{\text{barrier}}\right)} \tag{6.3}$$

[1] ρ is used to indicate both resistivity and best stage effort. The meaning should be clear from context.

Example 6.1

Compute the sheet resistance of a 0.22 μm thick Cu wire in a 65 nm process. Find the total resistance if the wire is 0.125 μm wide and 1 mm long. Ignore the barrier layer and dishing.

SOLUTION: The sheet resistance is

$$R_{\square} = \frac{2.2 \times 10^{-8}\ \Omega \cdot \text{m}}{0.22 \times 10^{-6}\ \text{m}} = 0.10\ \Omega/\square \tag{6.4}$$

The total resistance is

$$R = (0.10\ \Omega/\square)\frac{1000\ \mu\text{m}}{0.125\ \mu\text{m}} = 800\ \Omega \tag{6.5}$$

The resistivity of polysilicon, diffusion, and wells is significantly influenced by the doping levels. Polysilicon and diffusion typically have sheet resistances under 10 Ω/square when silicided and up to several hundred Ω/square when unsilicided. Wells have lower doping and thus even higher sheet resistance. These numbers are highly process-dependent. Large resistors are often made from wells or unsilicided polysilicon.

Contacts and vias also have a resistance, which is dependent on the contacted materials and size of the contact. Typical values are 2–20 Ω. Multiple contacts should be used to form low-resistance connections, as shown in Figure 6.8. When current turns at a right angle or reverses, a square array of contacts is generally required, while fewer contacts can be used when the flow is in the same direction.

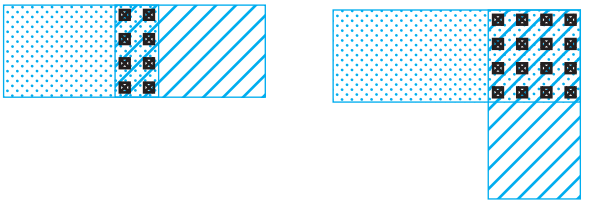


FIGURE 6.8 Multiple vias for low-resistance connections

6.2.2 Capacitance

An isolated wire over the substrate can be modeled as a conductor over a ground plane. The wire capacitance has two major components: the parallel plate capacitance of the bottom of the wire to ground and the fringing capacitance arising from fringing fields along the edge of a conductor with finite thickness. In addition, a wire adjacent to a second wire on the same layer can exhibit capacitance to that neighbor. These effects are illustrated in Figure 6.9. The classic parallel plate capacitance formula is

$$C = \frac{\varepsilon_{ox}}{h} wl \tag{6.6}$$

Note that oxides are often doped with phosphorous to trap ions before they damage transistors; this oxide has $\varepsilon_{ox} \approx k\varepsilon_0$, with $k = 4.1$ as compared to 3.9 for an ideal oxide or lower for low-k dielectrics.

The fringing capacitance is more complicated to compute and requires a numerical field solver for exact results. A number of authors have proposed approximations to this calculation [Barke88, Ruehli73, Yuan82]. One intuitively

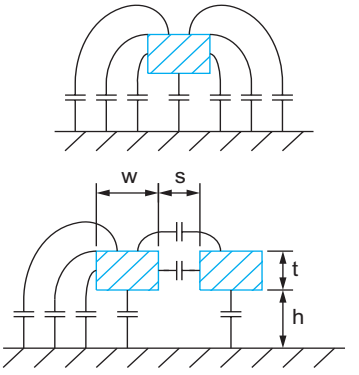


FIGURE 6.9 Effect of fringing fields on capacitance

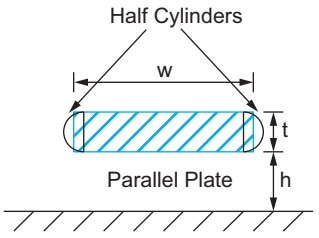


FIGURE 6.10 Yuan & Trick capacitance model including fringing fields

appealing approximation treats a lone conductor above a ground plane as a rectangular middle section with two hemispherical end caps, as shown in Figure 6.10 [Yuan82]. The total capacitance is assumed to be the sum of a parallel plate capacitor of width $w - t/2$ and a cylindrical capacitor of radius $t/2$. This results in an expression for the capacitance that is accurate within 10% for aspect ratios less than 2 and $t \approx h$.

$$C = \varepsilon_{ox} l \left[\frac{w - \frac{t}{2}}{h} + \frac{2\pi}{\ln\left(1 + \frac{2h}{t} + \sqrt{\frac{2h}{t}\left(\frac{2h}{t} + 2\right)}\right)} \right] \tag{6.7}$$

An empirical formula that is computationally efficient and relatively accurate is [Meijs84, Barke88]

$$C = \varepsilon_{ox} l \left[\frac{w}{h} + 0.77 + 1.06\left(\frac{w}{h}\right)^{0.25} + 1.06\left(\frac{t}{h}\right)^{0.5} \right] \tag{6.8}$$

which is good to 6% for aspect ratios less than 3.3.

These formulae do not account for neighbors on the same layer or higher layers. Capacitance interactions between layers can become quite complex in modern multilayer CMOS processes. A conservative upper bound on capacitance can be obtained assuming parallel neighbors on the same layer at minimum spacing and that the layers above and below the conductor of interest are solid ground planes. Similarly, a lower bound can be obtained assuming there are no other conductors in the system except the substrate. The upper bound can be used for propagation delay and power estimation while the lower bound can be used for contamination delay calculations before layout information is available.

A cross-section of the model used for capacitance upper bound calculations is shown in Figure 6.11. The total capacitance of the conductor of interest is the sum of its capacitance to the layer above, the layer below, and the two adjacent conductors. If the layers above and below are not switching,[2] they can be modeled as ground planes and this component of capacitance is called C_{gnd}. Wires do have some capacitance to further neighbors, but this capacitance is generally negligible because most electric fields terminate on the nearest conductors. The dielectrics used between adjacent wires have the lowest possible dielectric constant k_{horiz} to minimize capacitance. The dielectric between layers must provide greater mechanical stability and may have a larger k_{vert}. EQ (6.9) gives a simple and physically intuitive estimate of wire capacitance [Bohr95]. The constant C_{fringe} term accounts for fringing capacitance and gives a better fit for w and s up to several times minimum [Ho01].

$$\begin{aligned} C_{total} &= C_{top} + C_{bot} + 2C_{adj} \\ &\approx \varepsilon_0 l \left[2k_{vert}\frac{w}{h} + 2k_{horiz}\frac{t}{s} \right] + C_{fringe} \end{aligned} \tag{6.9}$$

[2] Or at least consist of a large number of orthogonal conductors that on average cancel each other's switching activities.

The capacitances can be computed by generating a lookup table of data with a field solver such as FastCap [Nabors92] or HSPICE. The table may contain data for different widths and spacings for each metal layer, assuming the layers above and below are occupied or unoccupied. The table should list both C_{adj} and C_{gnd}, because coupling to adjacent lines is of great importance. Figure 6.12 shows representative data for a metal2 wire in a 180 nm process with wire and oxide thicknesses of 0.7 μm. The width and spacing are given in multiples of the 0.32 μm minimum. For an isolated wire above the substrate, the capacitance is strongly influenced by spacing between conductors. For a wire sandwiched between metal1 and metal3 planes, the capacitance is higher and is more sensitive to the width (determining parallel plate capacitance) but less sensitive to spacing once the spacing is significantly greater than the wire thickness. In either case, the y-intercept is greater than zero so doubling the width of a wire results in less than double the total capacitance. The data fits EQ (6.9) with C_{fringe} = 0.05 fF/μm. Tight-pitch metal lines have a capacitance of roughly 0.2 fF/μm.

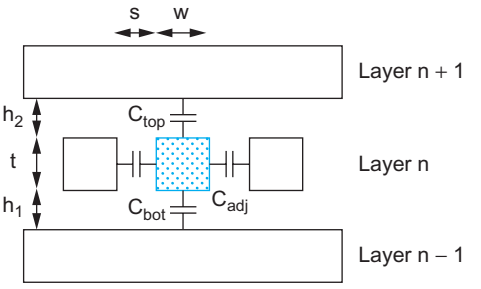


FIGURE 6.11 Multilayer capacitance model

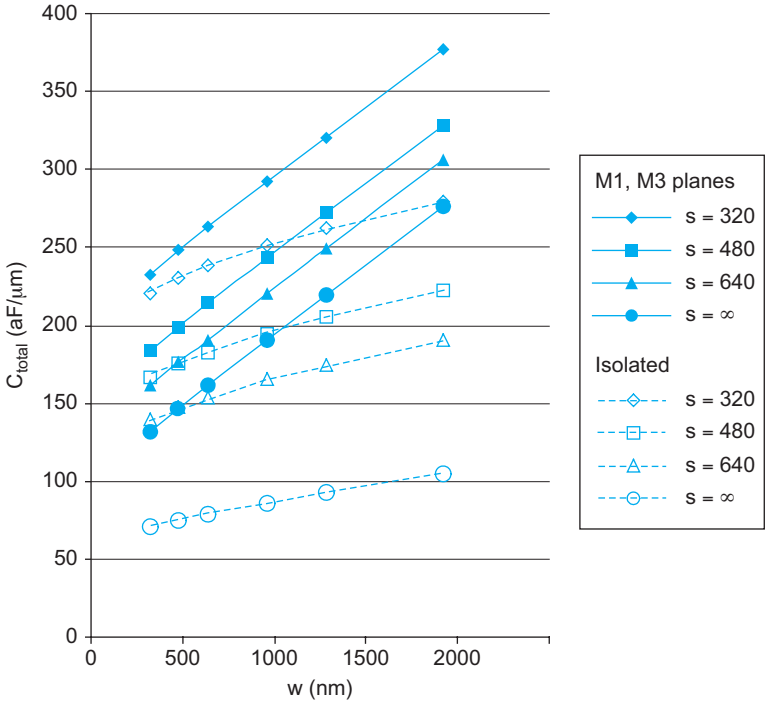


FIGURE 6.12 Capacitance of metal2 line as a function of width and spacing

In practice, the layers above and below the conductor of interest are neither solid planes nor totally empty. One can extract capacitance more accurately by interpolating between these two extremes based on the density of metal on each level. [Chern92] gives formulae for this interpolation accurate to within 10%. However, if the wiring above and below is fairly dense (e.g., a bus on minimum pitch), it is well-approximated as a plane. Dense wire fill is added to many chips for mechanical stability and etch uniformity, making this approximation even more appropriate.

6.2.3 Inductance

Most design tools consider only interconnect resistance and capacitance. Inductance is difficult to extract and model, so engineers prefer to design in such a way that inductive effects are negligible. Nevertheless, inductance needs to be considered in high-speed designs for wide wires such as clocks and power busses.

Although we generally discuss current flowing from a gate output to charge or discharge a load capacitance, current really flows in loops. The return path for a current loop is usually the power or ground network; at the frequencies of interest, the power supply is an "AC ground" because the bypass capacitance forms a low-impedance path between V_{DD} and GND. Currents flowing around a loop generate a magnetic field proportional to the area of the loop and the amount of current. Changing the current requires supplying energy to change the magnetic field. This means that changing currents induce a voltage proportional to the rate of change. The constant of proportionality is called the inductance, L.[3]

$$V = L\frac{dI}{dt} \tag{6.10}$$

Inductance and capacitance also set the speed of light in a medium. Even if the resistance of a wire is zero leading to zero RC delay, the speed of light flight-time along a wire of length with inductance and capacitance per unit length of L and C is

$$t_{pd} = l\sqrt{LC} \tag{6.11}$$

If the current return paths are the same as the conductors on which electric field lines terminate, the signal velocity v is

$$v = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\varepsilon_{ox}\mu_0}} = \frac{c}{\sqrt{3.9}} \tag{6.12}$$

where μ_0 is the magnetic permeability of free space ($4\pi \times 10^{-7}$ H/m) and c is the speed of light in free space (3×10^8 m/s). In other words, signals travel about half the speed of light. Using low-k (< 3.9) dielectrics raises this velocity. However, many signals have electric fields terminating on nearby neighbors, but currents returning in more distant power supply lines. This raises the inductance and reduces the signal velocity.

Changing magnetic fields in turn produce currents in other loops. Hence, signals on one wire can inductively couple onto another; this is called *inductive crosstalk*.

The inductance of a conductor of length l and width w located a height h above a ground plane is approximately

$$L = l\frac{\mu_0}{2\pi}\ln\left(\frac{8h}{w} + \frac{w}{4h}\right) \tag{6.13}$$

assuming $w < h$ and thickness is negligible. Typical on-chip inductance values are in the range of 0.15–1.5 pH/μm depending on the proximity of the power or ground lines. (Wires near their return path have smaller current loops and lower inductance.)

[3] L is used to indicate both inductance and transistor channel length. The meaning should be clear from context.

Extracting inductance in general is a three-dimensional problem and is extremely time-consuming for complex geometries. Inductance depends on the entire loop and therefore cannot be simply decomposed into sections as with capacitance. It is therefore impractical to extract the inductance from a chip layout. Instead, usually inductance is extracted using tools such as FastHenry [Kamon94] for simple test structures intended to capture the worst cases on the chip. This extraction is only possible when the power supply network is highly regular. Power planes are ideal but require a large amount of metal resources. Dense power grids are usually the preferred alternative. Gaps in the power grid force current to flow around the gap, increasing the loop area and greatly increasing inductance. Moreover, large loops couple magnetic fields through other loops formed by conductors at a distance. Therefore, mutual inductive coupling can occur over a long distance, especially when the return path is far from the conductor.

6.2.4 Skin Effect

Current flows along the path of lowest impedance $Z = R + j\omega L$. At high frequency, ω, impedance becomes dominated by inductance. The inductance is minimized if the current flows only near the surface of the conductor closest to the return path(s). This skin effect can reduce the effective cross-sectional area of thick conductors and raise the effective resistance at high frequency. The skin depth for a conductor is

$$\delta = \sqrt{\frac{2\rho}{\omega\mu}} \tag{6.14}$$

where μ is the magnetic permeability of the dielectric (normally the same as in free space, $4\pi \times 10^{-7}$ H/m). The frequency of importance is the highest frequency with significant power in the Fourier transform of the signal. This is not the chip operating frequency, but rather is associated with the faster edges. A sine wave with the same 20–80% rise/fall time as the signal has a period of $8.65t_{rf}$. Therefore, the frequency associated with the edge can be approximated as

$$\omega = \frac{2\pi}{8.65\, t_{rf}} \tag{6.15}$$

where t_{rf} is the average 20–80% rise/fall time.

In a chip with a good power grid, good current return paths are usually available on all sides. Thus, it is a reasonable approximation to assume the current flows in a shell of thickness δ along the four sides of the conductor, as shown in Figure 6.13. If $\min(w, t) > 2\delta$, part of the conductor carries no current and the resistance increases.

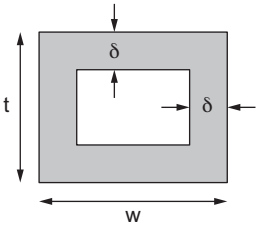


FIGURE 6.13 Current flow in shell determined by skin depth

Example 6.2

Determine the skin depth for a copper wire in a chip with 20 ps edge rates.

SOLUTION: According to EQ (6.15), the maximum frequency of interest is

$$\omega = \frac{2\pi}{8.65 \times 20\ \text{ps}} = 3.6 \times 10^{10}\ \text{rad/s} = 5.8\ \text{GHz} \tag{6.16}$$

According to EQ (6.14), the skin depth is thus

$$\delta = \sqrt{\frac{2\left(2.2\times10^{-8}\ \Omega\cdot\mathrm{m}\right)}{\left(3.6\times10^{10}\ \mathrm{rad/s}\right)\left(4\pi\times10^{-7}\ \mathrm{H/m}\right)}} = 0.99\ \mu\mathrm{m} \tag{6.17}$$

This exceeds half the thickness of typical metal layers, so the skin effect is rarely a factor in CMOS circuits.



6.2.5 Temperature Dependence

Interconnect capacitance is independent of temperature, but the resistance varies strongly. The temperature coefficients of copper and aluminum are about 0.4%/°C over the normal operating range of circuits; that is, a 100 °C increase in temperature leads to 40% higher resistance. At liquid nitrogen temperature (77 K), the resistivity of copper drops to 0.22 $\mu\Omega \cdot$ cm, an order-of-magnitude improvement. This suggests great advantages for RC-dominated paths in cooled systems.

6.3 Interconnect Impact

Using the lumped models, this section examines the delay, energy, and noise impact of wires.

6.3.1 Delay

Interconnect increases circuit delay for two reasons. First, the wire capacitance adds loading to each gate. Second, long wires have significant resistance that contributes distributed RC delay or *flight time*. It is straightforward to add wire capacitance to the Elmore delay calculations of Section 4.3.5, so in this section we focus on the RC delay.

The Elmore delay of a single-segment L-model is *RC*. As the number of segments of the L-model increases, the Elmore delay decreases toward *RC*/2. The Elmore delay of a π- or T-model is *RC*/2 no matter how many segments are used. Thus, a single-segment π-model is a good approximation for hand calculations.

Example 6.3

A 10x unit-sized inverter drives a 2x inverter at the end of the 1 mm wire from Example 6.1. Suppose that wire capacitance is 0.2 fF/μm and that unit-sized nMOS transistor has R = 10 kΩ and C = 0.1 fF. Estimate the propagation delay using the Elmore delay model; neglect diffusion capacitance.

SOLUTION: The driver has a resistance of 1 kΩ.The receiver has a 2-unit nMOS transistor and a 4-unit pMOS transistor, for a capacitance of 0.6 fF. The wire capacitance is 200 fF.

Figure 6.14 shows an equivalent circuit for the system using a single-segment π-model. The Elmore delay is t_{pd} = (1000 Ω)(100 fF) + (1000 Ω + 800 Ω)(100 fF + 0.6 fF) = 281 ps. The capacitance of the long wire dominates the delay; the capacitance of the 2x inverter is negligible in comparison.

Because both wire resistance and wire capacitance increase with length, wire delay grows quadratically with length. Using thicker and wider wires, lower-resistance metals such as copper, and lower-dielectric constant insulators helps, but long wires nevertheless often have unacceptable delay. Section 6.4.2 describes how repeaters can be used to break a long wire into multiple segments such that the overall delay becomes a linear function of length.

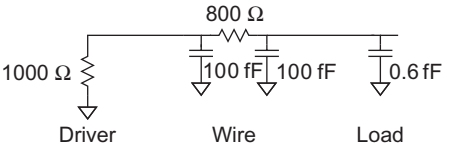


FIGURE 6.14 Equivalent circuit for example

Example 6.4

Find the RC flight time per mm^2 for a wire using the parameters from Example 6.3. Express the result in FO4/mm^2, if the FO4 inverter delay is 15 ps. What is the flight time to cross a 10 mm die?

SOLUTION: $R = 800\ \Omega$/mm. $C = 0.2$ pF/mm. The flight time is $RC/2 = 80$ ps/mm^2, or 5.3 FO4/mm^2. The flight time across a 10 mm die is thus 530 FO4, which is dozens of clock cycles.

Polysilicon and diffusion wires (sometimes called *runners*) have high resistance, even if silicided. Diffusion also has very high capacitance. Do not use diffusion for routing. Use polysilicon sparingly, usually in latches and flip-flops (i.e., do not use for other than intra-cell routing).

Recall that the Elmore delay model only considers the resistance on the path from the driver to a leaf. Capacitances on other branches are lumped as if they were at the branch point. This gives a conservative result because they are really partially shielded by their resistances.

Example 6.5

Figure 6.15 models a gate driving wires to two destinations. The gate is represented as a voltage source with effective resistance R_1. The two receivers are located at nodes 3 and 4. The wire to node 3 is long enough that it is represented with a pair of π-segments, while the wire to node 4 is represented with a single segment. Find the Elmore delay from input x to each receiver.

SOLUTION: The Elmore delays are

$$T_{D_3} = R_1C_1 + (R_1 + R_2)C_2 + (R_1 + R_2 + R_3)C_3 + R_1C_4$$
$$T_{D_4} = R_1C_1 + R_1(C_2 + C_3) + (R_1 + R_4)C_4 \qquad (6.18)$$

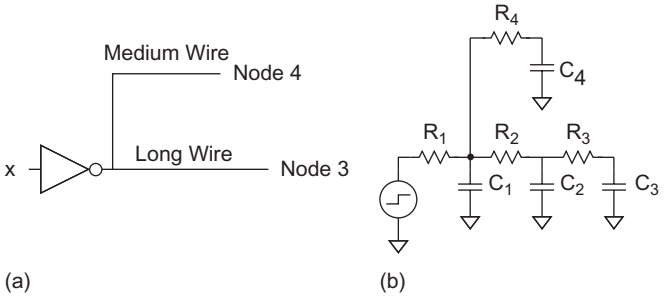


FIGURE 6.15 Interconnect modeling with RC tree

6.3.2 Energy

The switching energy of a wire is set by its capacitance. Long wires have significant capacitance and thus require substantial amounts of energy to switch.

Example 6.6

Estimate the energy per unit length to send a bit of information (one rising and one falling transition) in a CMOS process.

SOLUTION: $E = (0.2\text{ pF/mm})\ (1.0\text{ V})^2 = 0.2\text{ pJ/bit/mm}$. Sometimes energy in a communication link is expressed as power per gigabit per second: 0.2 mW/Gbps.

Example 6.7

Consider a microprocessor on a 20 mm × 20 mm die running at 3 GHz in the 65 nm process. A layer of metal is routed on a 250 nm pitch. Half of the available wire tracks are used. The wires have an average activity factor of 0.1. Determine the power consumed by the layer of metal.

SOLUTION: There are (20 mm) / (250 nm) = 80,000 tracks of metal across the die, of which 40,000 are occupied. The wire capacitance is (0.2 pF/mm)(20 mm)(40,000 tracks) = 160 nF. The power is $(0.1)(160\text{ nF})(1.0\text{ V})^2(3\text{ GHz}) = 48\text{ W}$. This is clearly a problem, especially considering that the chip has more than one layer of metal. The activity factor needs to be much lower to keep power under control.

6.3.3 Crosstalk

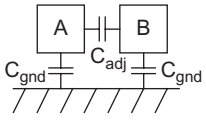


FIGURE 6.16 Capacitances to adjacent neighbor and to ground

As reviewed in Figure 6.16, wires have capacitance to their adjacent neighbors as well as to ground. When wire *A* switches, it tends to bring its neighbor *B* along with it on account of capacitive coupling, also called *crosstalk*. If *B* is supposed to switch simultaneously, this may increase or decrease the switching delay. If *B* is not supposed to switch, crosstalk causes noise on *B*. We will see that the impact of crosstalk depends on the ratio of C_{adj} to the total capacitance. Note that the load capacitance is included in the total, so for short wires and large loads, the load capacitance dominates and crosstalk is unimportant. Conversely, crosstalk is very important for long wires.

6.3.3.1 Crosstalk Delay Effects If both a wire and its neighbor are switching, the direction of the switching affects the amount of charge that must be delivered and the delay of the switching. Table 6.3 summarizes this effect. The charge delivered to the coupling capacitor is $Q = C_{adj}\Delta V$, where ΔV is the change in voltage between *A* and *B*. If *A* switches but *B* does not, $\Delta V = V_{DD}$. The total capacitance effectively seen by *A* is just the capacitance to ground and to *B*. If both *A* and *B* switch in the same direction, $\Delta V = 0$. Hence, no charge is required and C_{adj} is effectively absent for delay purposes. If *A* and *B* switch in the opposite direction, $\Delta V = 2V_{DD}$. Twice as much charge is required. Equivalently, the capacitor can be treated as being effectively twice as large switching through V_{DD}. This is analogous to the Miller effect discussed in Section 4.4.6.6. The *Miller Coupling Factor* (MCF) describes how the capacitance to adjacent wires is multiplied to find the effective capacitance. Some designers use MCF = 1.5 as a statistical compromise when estimating propagation delays before layout information is available.

TABLE 6.3 Dependence of effective capacitance on switching direction

B	ΔV	$C_{eff(A)}$	MCF
Constant	V_{DD}	$C_{gnd} + C_{adj}$	1
Switching same direction as *A*	0	C_{gnd}	0
Switching opposite to *A*	$2V_{DD}$	$C_{gnd} + 2C_{adj}$	2

A conservative design methodology assumes neighbors are switching when computing propagation and contamination delays (MCF = 2 and 0, respectively). This leads to a wide variation in the delay of wires. A more aggressive methodology tracks the time window during which each signal can switch. Thus, switching neighbors must be accounted for only if the potential switching windows overlap. Similarly, the direction of switching can be considered. For example, dynamic gates described in Section 9.2.4 precharge high and then fall low during evaluation. Thus, a dynamic bus will never see opposite switching during evaluation.

Example 6.8

Each wire in a pair of 1 mm lines has capacitance of 0.08 fF/μm to ground and 0.12 fF/μm to its neighbor. Each line is driven by an inverter with a 1 kΩ effective resistance. Estimate the contamination and propagation delays of the path. Neglect parasitic capacitance of the inverter and resistance of the wires.

SOLUTION: We find C_{gnd} = (0.08 fF/μm)(1000 μm) = 80 fF and C_{adj} = 120 fF. The delay is RC_{eff}. The contamination delay is the minimum possible delay, which occurs when both wires switch in the same direction. In that case, $C_{eff} = C_{gnd}$ and the delay is t_{cd} = (1 kΩ)(0.08 pF) = 80 ps. The propagation delay is the maximum possible delay, which occurs when both wires switch in opposite directions. In this case, $C_{eff} = C_{gnd} + 2C_{adj}$ and the delay is t_{pd} = (1 kΩ)(0.32 pF) = 320 ps. This is a factor of four difference between best and worst case.

6.3.3.2 Crosstalk Noise Effects Suppose wire *A* switches while *B* is supposed to remain constant. This introduces noise as *B* partially switches. We call *A* the *aggressor* or *perpetrator* and *B* the *victim*. If the victim is floating, we can model the circuit as a capacitive voltage divider to compute the victim noise, as shown in Figure 6.17. $\Delta V_{aggressor}$ is normally V_{DD}.

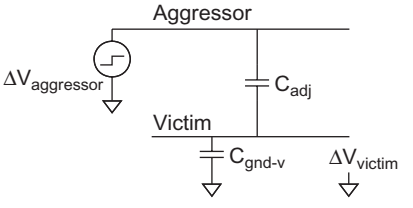


FIGURE 6.17 Coupling to floating victim

$$\Delta V_{victim} = \frac{C_{adj}}{C_{gnd-v} + C_{adj}} \Delta V_{aggressor} \quad (6.19)$$

If the victim is actively driven, the driver will supply current to oppose and reduce the victim noise. We model the drivers as resistors, as shown in Figure 6.18. The peak noise becomes dependent on the time constant ratio *k* of the aggressor to the victim [Ho01]:

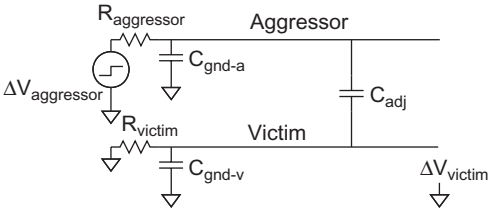


FIGURE 6.18 Coupling to driven victim

$$\Delta V_{victim} = \frac{C_{adj}}{C_{gnd-v} + C_{adj}} \frac{1}{1+k} \Delta V_{aggressor} \quad (6.20)$$

where

$$k = \frac{\tau_{\text{aggressor}}}{\tau_{\text{victim}}} = \frac{R_{\text{aggressor}}\left(C_{\text{gnd}-a} + C_{\text{adj}}\right)}{R_{\text{victim}}\left(C_{\text{gnd}-v} + C_{\text{adj}}\right)} \tag{6.21}$$

Figure 6.19 shows simulations of coupling when the aggressor is driven with a unit inverter; the victim is undriven or driven with an inverter of half, equal, or twice the size of the aggressor; and $C_{\text{adj}} = C_{\text{gnd}}$. Observe that when the victim is floating, the noise remains indefinitely. When the victim is driven, the driver restores the victim. Larger (faster) drivers oppose the coupling sooner and result in noise that is a smaller percentage of the supply voltage. Note that during the noise event the victim transistor is in its linear region while the aggressor is in saturation. For equal-sized drivers, this means $R_{\text{aggressor}}$ is two to four times R_{victim}, with greater ratios arising from more velocity saturation [Ho01]. In general, EQ (6.20) is conservative, especially when wire resistance is included [Vittal99]. It is often used to flag nets where coupling can be a problem; then simulations can be performed to calculate the exact coupling noise. Coupling noise is of greatest importance on weakly driven nodes where $k < 1$.

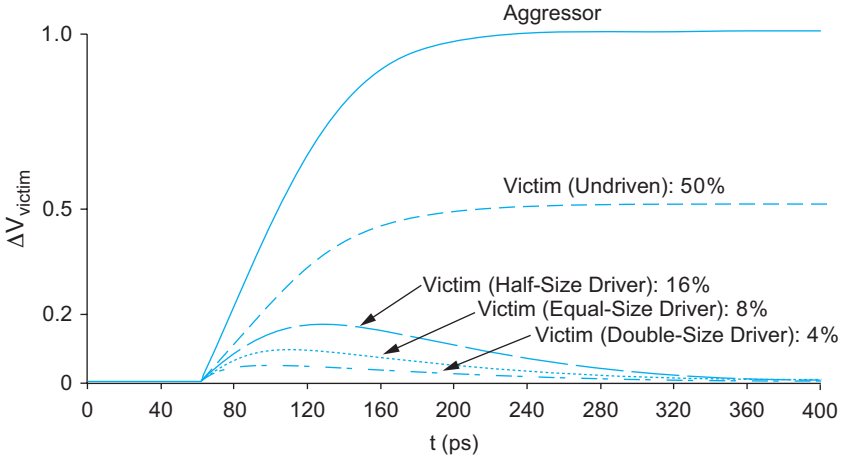


FIGURE 6.19 Waveforms of coupling noise

We have only considered the case of a single neighbor switching. When both neighbors switch, the noise will be twice as great. We have also modeled the layers above and below as AC ground planes, but wires on these layers are likely to be switching. For a long line, you can expect about as many lines switching up and switching down, giving no net contribution to delay or noise. However, a short line running over a 64-bit bus in which all 64 bits are simultaneously switching from 0 to 1 will be strongly influenced by this switching.

OPTIONAL

6.3.4 Inductive Effects

Inductance has always been important for integrated circuit packages where the physical dimensions are large, as will be discussed in Section 13.2.3. On-chip inductance is important for wires where the speed-of-light flight time is longer than either the rise times of the circuits or the RC delay of the wire. Because speed-of-light flight time increases lin-

early according to EQ (6.11) and RC delay increases quadratically with length, we can estimate the set of wire lengths for which inductance is relevant [Ismail99].

$$\frac{t_r}{2\sqrt{LC}} < l < \frac{2}{R}\sqrt{\frac{L}{C}} \tag{6.22}$$

Example 6.9

Consider a metal2 signal line with a sheet resistance of 0.10 Ω/□ and a width of 0.125 μm. The capacitance is 0.2 fF/μm and inductance is 0.5 pH/μm. Compute the velocity of signals on the line and plot the range of lengths over which inductance matters as a function of the rise time.

SOLUTION: The velocity is

$$v = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.5\ \text{pH}/\mu\text{m})(0.2\ \text{fF}/\mu\text{m})}} = 10^8\ \text{m/s} = \frac{1}{3}c \tag{6.23}$$

Note that this is 100 mm/ns or 1 mm/10 ps. The resistance is (0.1 Ω/□)(1 □/0.125 μm) = 0.8 Ω/μm. Figure 6.20 plots the length of wires for which inductance is relevant against rise times. Above the horizontal line, wires greater than 125 μm are limited by RC delay rather than LC delay. To the right of the diagonal line, rise times are greater than the LC delay. Only in the region between these lines is inductance relevant to delay calculations. This region has very fast edge rates, so inductance is not very important to the delay of highly resistive signal lines.

As the example illustrated, inductance will only be important to the delay of low-resistance signals such as wide clock lines. Inductive crosstalk is also important for wide busses far away from their current return paths. In power distribution networks, inductance means that if one portion of the chip requires a rapidly increasing amount of current, that charge must be delivered from nearby decoupling capacitors or supply pins; portions of the chip further away are unaware of the changing current needs until a speed-of-light flight time has elapsed and hence will not supply current immediately. Adding inductance to the power grid simulation generally reveals greater supply noise than would otherwise be predicted. Power networks will be discussed further in Section 13.3.

In wide, thick, upper-level metal lines, resistance and RC delay may be small. This pushes the horizontal line in Figure 6.20 upward, increasing the range of edge rates for which inductance matters. This is especially common for clock signals. Inductance tends to increase the propagation delay and sharpen the edge rate.

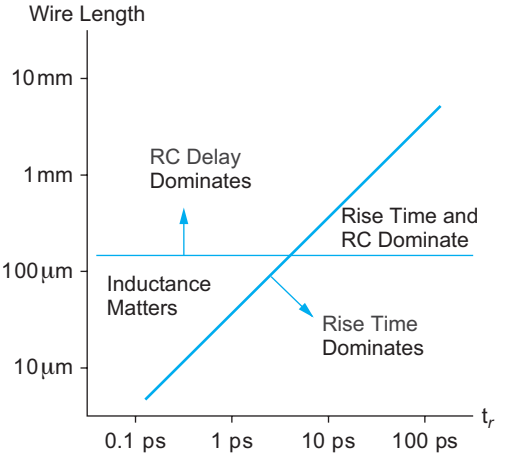


FIGURE 6.20 Wire lengths and edge rates for which inductance impacts delay

To see the effects of inductance, consider a 5 mm-long clock line above a ground plane driving a 2 pF clock load. If its width is 4.8 μm and thickness is 1.7 μm, it has resistance of 4 Ω/mm, capacitance of 0.4 pF/mm, and inductance of 0.12 nH/mm. Figure 6.21 presents models of the clock line as a 5-stage π-model without (a) and with (b) inductance. Figure 6.21(c) shows the

response of each model to an ideal voltage source with 80 ps rise time. The model including inductance shows a greater delay until the clock begins to rise because of the speed-of-light flight time. It also overshoots. However, the rising edge is sharper and the rise time is shorter. In some circumstances when the driver impedance is matched to the characteristic impedance of the wire, the sharper rising edge can actually result in a shorter propagation delay measured at the 50% point.

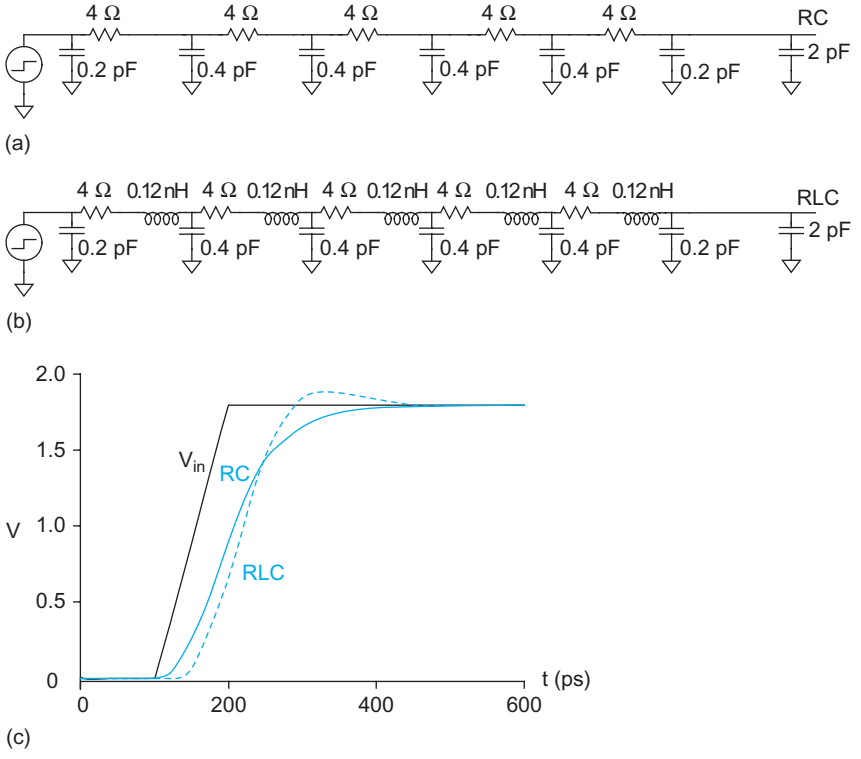


FIGURE 6.21 Wide clock line modeled with and without inductance

To reduce the inductance and the impact of skin effect when no ground plane is available, it is good practice to split wide wires into thinner sections interdigitated with power and ground lines to serve as return paths. For example, Figure 6.22 shows how a 16 μm-wide clock line can be split into four 4 μm lines to reduce the inductance.

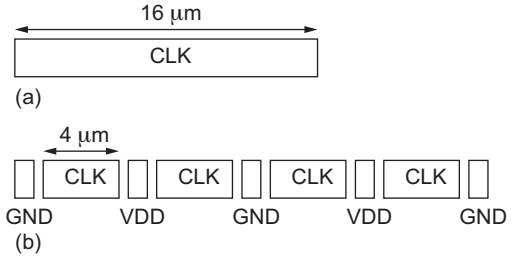


FIGURE 6.22 Wide clock line interdigitated with power and ground lines to reduce inductance

A bus made of closely spaced wires far above a ground plane is particularly susceptible to crosstalk. Figure 6.23 shows the worst case crosstalk scenario. The victim line is in the center. The two adjacent neighbors rise, capacitively coupling the victim upward. The other bus wires fall. Each one creates a loop of current flowing counterclockwise through the wire and back along the ground plane. These loops induce a magnetic field, which in turn induces a current flowing in the other direction in the victim line. This is called mutual inductive coupling and also makes the victim rise. The noise from each aggressor sums on to the victim in much the same way that multiple primary turns in a transformer couple onto a single secondary turn. Computing the inductive crosstalk requires extracting a mutual inductance matrix for the bus and simulating the system. As this is not yet practical for large chips, designers instead either follow design rules that keep the inductive effects small or ignore inductance and hope for the best. The design rules may be of the form that one power or ground wire must be inserted between every N signal lines on each layer. N is called the signal:return (SR) ratio [Morton99]. The returns give an alternative path for current to flow, reducing the mutual inductance. The inductive effects on noise and delay are generally small for $N = 8$ and negligible for $N = 4$ when normal wiring pitches are used [Linderman04]. $N = 2$ means each signal is shielded on one side, also eliminating half the capacitive crosstalk. However, low SR ratios are expensive in terms of metal resources.

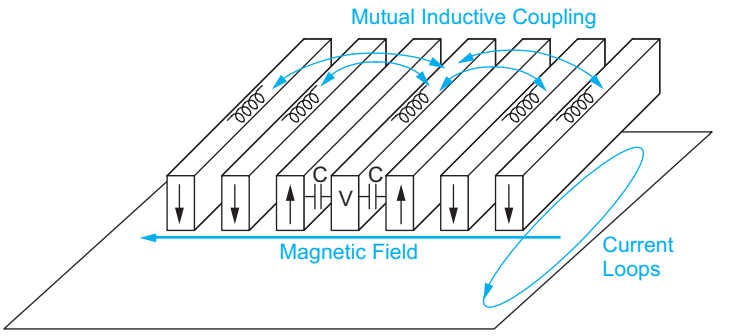


FIGURE 6.23 Inductive and capacitive crosstalk in a bus

In summary, on-chip inductance is difficult to extract. Mutual inductive coupling may occur over a long range, so inductive coupling is difficult to simulate even if accurate values are extracted. Instead, design rules are usually constructed so that inductive effects may be neglected for most structures. The easiest way to do this is to provide a regular power grid in which power and ground are systematically allocated track to keep the SR ratio low. Inductance should be incorporated into simulations of the power and clock networks and into the noise and delay calculations for busses with large SR ratios in high-speed designs.

6.3.5 An Aside on Effective Resistance and Elmore Delay

OPTIONAL

Recall from Section 4.3.4 that a factor of ln 2 was lumped into the effective resistance of a transistor so that the Elmore delay model predicts propagation delay, yet we have not accounted for the factor in wire resistance. This section examines the discrepancy.

According to the Elmore delay model, a gate with effective resistance R and capacitance C has a propagation delay of RC. A wire with distributed resistance R and capacitance C treated as a single π-segment has propagation delay $RC/2$. Reviewing the properties of RC circuits, we recall that the lumped RC circuit in Figure 6.24(a) has a unit step response of

$$V_{out}(t) = 1 - e^{\frac{-t}{RC}} \tag{6.24}$$

The propagation delay of this circuit is obtained by solving for t_{pd} when $V_{out}(t_{pd}) = 1/2$:

$$t_{pd} = R'C \ln 2 = 0.69R'C \tag{6.25}$$

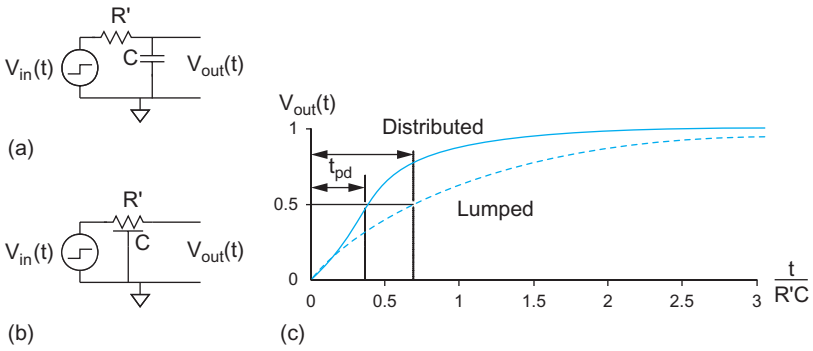


FIGURE 6.24 Lumped and distributed RC circuit response

The distributed RC circuit in Figure 6.24(b) has no closed form time domain response. Because the capacitance is distributed along the circuit rather than all being at the end, you would expect the capacitance to be charged on average through about half the resistance and that the propagation delay should thus be about half as great. A numerical analysis finds that the propagation delay is $0.38R'C$.

To reconcile the Elmore model with the true results for a logic gate, recall that logic gates have complex nonlinear I-V characteristics and are approximated as having an effective resistance. If we characterize that effective resistance as $R = R' \ln 2$, the propagation delay really becomes the product of the effective resistance and the capacitance: $t_{pd} = RC$.

For distributed circuits, observe that

$$0.38R'C \approx \frac{1}{2}R'C \ln 2 = \frac{1}{2}RC$$

Therefore, the Elmore delay model describes distributed delay well if we use an effective wire resistance scaled by ln 2 from that computed with EQ (6.2). This is somewhat inconvenient. The effective resistance is further complicated by the effect of nonzero rise time on propagation delay. Figure 6.25 shows that the propagation delay depends on the rise time of the input and approaches RC for lumped systems and $RC/2$ for distributed systems when the input is a slow ramp. This suggests that when the input is slow, the effective resistance for delay calculations in a distributed RC circuit is equal to the true resistance. Finally, we note that for many analyses such as repeater insertion calculations in Section 6.4.2, the results are only weakly sensitive to wire resistance, so using the true wire resistance does not introduce great error.

In summary, it is a reasonable practice to estimate the flight time along a wire as $RC/2$ where R is the true resistance of the wire. When more accurate results are needed, it is important to use good transistor models and appropriate input slopes in simulation.

The Elmore delay can be viewed in terms of the first moment of the impulse response of the circuit. CAD tools can obtain greater accuracy by approximating delay based on higher moments using a technique called *moment matching*. *Asymptotic Waveform Evaluation* (AWE) uses moment matching to estimate interconnect delay with better accuracy than the Elmore delay model and faster run times than a full circuit simulation [Celik02].

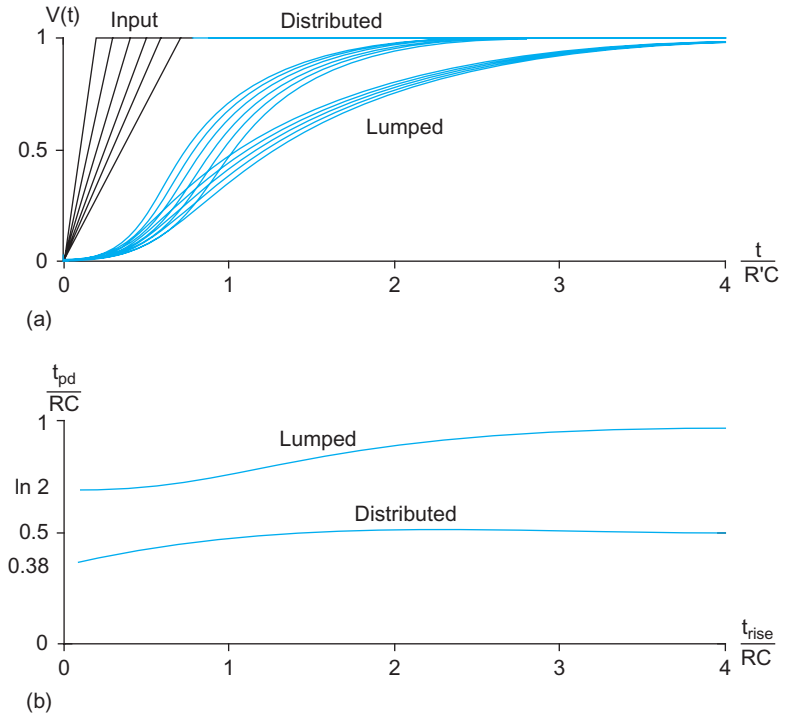


FIGURE 6.25 Effect of rise time on lumped and distributed RC circuit delays

6.4 Interconnect Engineering

As gate delays continue to improve while long wire delays remain constant or even get slower, wire engineering has become a major part of integrated circuit design. It is necessary to develop a floorplan early in the design cycle, identify the long wires, and plan for them. While floorplanning in such a way that critical communicating units are close to one another has the greatest impact on performance, it is inevitable that long wires will still exist. Aspect ratios in old processes were below 1, but are close to 2 in nanometer processes to help the resistance of such narrow lines. This comes at the expense of substantially increased coupling capacitance. The designer has a number of techniques to engineer wires for delay and coupling noise. The width, spacing, and layer usage are all under the designer's control. Shielding can be used to further reduce coupling on critical nets. Repeaters inserted along long wires reduce the delay from a quadratic to a linear function of length. Wire capacitance and resistance complicate the use of Logical Effort in selecting gate sizes.

6.4.1 Width, Spacing, and Layer

The designer selects the wire width, spacing, and layer usage to trade off delay, bandwidth, energy, and noise. By default, minimum pitch wires are preferred for noncritical intercon-

nections for best density and bandwidth. When the load is dominated by wire capacitance, the best way to reduce delay is to increase spacing, reducing the capacitance to nearby neighbors. This also reduces energy and coupling noise. When the delay is dominated by the gate capacitance and wire resistance, widening the wire reduces resistance and delay. However, it increases the capacitance of the top and bottom plates. Widening wires also increases the fraction of capacitance of the top and bottom plates, which somewhat reduces coupling noise from adjacent wires. However, wider wires consume more energy.

The wire thickness depends on the choice of metal layer. The lower layers are thin and optimized for a tight routing pitch. Middle layers are often slightly thicker for lower resistance and better current-handling capability. Upper layers may be even thicker to provide a low-resistance power grid and fast global interconnect. Wiring tracks are a precious resource and are often allocated in the floorplan; the wise designer maintains a reserve of wiring tracks for unanticipated changes late in the design process.

The power grid is usually distributed over multiple layers. Most of the current-handling capability is provided in the upper two layers with lowest resistance. However, the grid must extend down to metal1 or metal2 to provide easy connection to cells.

6.4.2 Repeaters

Both resistance and capacitance increase with wire length l, so the RC delay of a wire increases with l^2, as shown in Figure 6.26(a). The delay may be reduced by splitting the wire into N segments and inserting an inverter or buffer called a *repeater* to actively drive the wire [Glasser85], as shown in Figure 6.26(b). The new wire involves N segments with RC flight time of $(l/N)^2$, for a total delay of l^2/N. If the number of segments is proportional to the length, the overall delay increases only linearly with l.

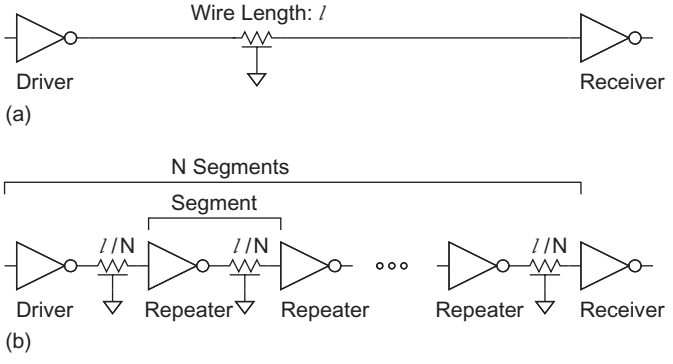


FIGURE 6.26 Wire with and without repeaters

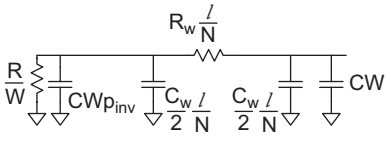


FIGURE 6.27 Equivalent circuit for segment of repeated wire

Using inverters as repeaters gives best performance. Each repeater adds some delay. If the distance is too great between repeaters, the delay will be dominated by the long wires. If the distance is too small, the delay will be dominated by the large number of inverters. As usual, the best distance between repeaters is a compromise between these extremes. Suppose a unit inverter has resistance R, gate capacitance C,[4] and diffusion capacitance Cp_{inv}. A wire has resistance R_w and capacitance C_w per unit length. Consider inserting repeaters of W times unit size.

[4]Note that C now refers to the capacitance of an entire inverter, not a single transistor, so $\tau = RC$.

Figure 6.27 shows a model of one segment. The Elmore delay of the repeated wire is

$$t_{pd} = N\left[\frac{R}{W}\left(C_w\frac{l}{N} + CW\left(1+p_{\text{inv}}\right)\right) + R_w\frac{l}{N}\left(\frac{C_w}{2}\frac{l}{N} + CW\right)\right] \tag{6.26}$$

Differentiating EQ (6.26) with respect to N and W shows that the best length of wire between repeaters is (see Exercise 6.5)

$$\frac{l}{N} = \sqrt{\frac{2RC\left(1+p_{\text{inv}}\right)}{R_w C_w}} \tag{6.27}$$

Recall from Example 4.10 that the delay of an FO4 inverter is $5RC$. Assuming $p_{\text{inv}} \approx 0.5$ using folded transistors, EQ (6.27) simplifies to

$$\frac{l}{N} = 0.77\sqrt{\frac{\text{FO4}}{R_w C_w}} \tag{6.28}$$

The delay per unit length of a properly repeated wire is

$$\frac{t_{pd}}{l} = \left(2+\sqrt{2\left(1+p_{\text{inv}}\right)}\right)\sqrt{RCR_w C_w} \approx 1.67\sqrt{\text{FO4}\, R_w C_w} \tag{6.29}$$

To achieve this delay, the inverters should use an nMOS transistor width of

$$W = \sqrt{\frac{RC_w}{R_w C}} \tag{6.30}$$

The energy per unit length to send a bit depends on the wire and repeater capacitances

$$\frac{E}{l} = C_w + NWC\left(1+p_{\text{inv}}\right) = C_w\left(1+\sqrt{\frac{1+p_{\text{inv}}}{2}}\right)V_{DD}^2 \approx 1.87C_w V_{DD}^2 \tag{6.31}$$

In other words, repeaters sized for minimum delay add 87% to the energy of an unrepeated wire.

Example 6.10

Compute the delay per mm of a repeated wire in a 65 nm process. Assume the wire is on a middle routing layer and has 2x width, spacing, and height, so its resistance is 200 Ω/mm and capacitance is 0.2 pF/mm. The FO4 inverter delay is 15 ps. Also find the repeater spacing and driver size to achieve this delay and the energy per bit.

SOLUTION: Using EQ (6.29), the delay is

$$t_{pd} = 1.67\sqrt{(15\text{ ps})(200\ \Omega/\text{mm})(0.2\text{ pF/mm})} = 41\text{ ps/mm} \tag{6.32}$$

This delay is achieved using a spacing of 0.45 mm between repeaters and an nMOS driver width of 18 μm (180x unit size). The energy per bit is 0.4 pJ/mm.

As one might expect, the curve of delay vs. distance and driver size is relatively flat near the minimum. Thus, substantial energy can be saved for a small increase in delay. At the minimum EDP point, the segments become 1.7x longer and the drivers are only 0.6x as large. The delay increases by 14% but the repeaters only add 30% to the energy of the unrepeated line [Ho01]. For the parameters in Example 6.10, the minimum EDP can be found numerically at a spacing of about 0.8 mm and a driver width of 11 μm (110x unit size), achieving an energy of 0.26 pJ/mm at a delay of 47 ps/mm. These longer segments are more susceptible to noise.

Unfortunately, inverting repeaters complicate design because you must either ensure an even number of repeaters on each wire or adapt the receiving logic to accept an inverted input. Some designers use inverter pairs (buffers) rather than single inverters to avoid the polarity problem. The pairs contribute more delay. However, the first inverter size W_1 may be smaller, presenting less load on the wire driving it. The second inverter may be larger, driving the next wire more strongly. You can show that the best size of the second inverter is $W_2 = kW_1$, where $k = 2.25$ if $p_{\text{inv}} = 0.5$. The distance between repeaters increases to (see Exercise 6.6)

$$\frac{l}{N} = \sqrt{\frac{2RC\left(k + \frac{1}{k} + 2p_{\text{inv}}\right)}{R_w C_w}} \approx 1.22\sqrt{\frac{\text{FO4}}{R_w C_w}} \tag{6.33}$$

The delay per unit length becomes

$$\frac{t_{pd}}{l} = 1.81\sqrt{\text{FO4}\, R_w C_w} \tag{6.34}$$

using transistor widths of

$$W_1 = \frac{W}{\sqrt{k}}, \quad W_2 = W\sqrt{k} \tag{6.35}$$

and the energy per bit per unit length is

$$\frac{E}{l} \approx 2.2 C_w V_{DD}^2 \tag{6.36}$$

This typically means that wires driven with noninverting repeaters are only about 8% slower per unit length than those using inverting repeaters. Only about two-thirds as many repeaters are required, simplifying floorplanning. Total repeater area and power increases slightly.

The overall delay is a weak function of the distance between repeaters, so it is reasonable to increase this distance to reduce the difficulty of finding places in the floorplan for repeaters while only slightly increasing delay. Repeaters impose directionality on a wire. Bidirectional busses and distributed tristate busses cannot use simple repeaters and hence are slower; this favors point-to-point unidirectional communications.



6.4.3 Crosstalk Control

Recall from EQ (6.20) that the capacitive crosstalk is proportional to the ratio of coupling capacitance to total capacitance. For modern wires with an aspect ratio (t/w) of 2 or

greater, the coupling capacitance can account for 2/3 to 3/4 of the total capacitance and crosstalk can create large amounts of noise and huge data-dependent delay variations. There are several approaches to controlling this crosstalk:

- Increase spacing to adjacent lines
- Shield wires
- Ensure neighbors switch at different times
- Crosstalk cancellation

The easiest approach to fix a minor crosstalk problem is to increase the spacing. If the crosstalk is severe, the spacing may have to be increased by more than one full track. In such a case, it is more efficient to shield critical signals with power or ground wires on one or both sides to eliminate coupling. For example, clock wires are usually shielded so that switching neighbors do not affect the delay of the clock wire and introduce clock jitter. Sensitive analog wires passing near digital signals should also be shielded.

An alternative to shielding is to interleave busses that are guaranteed to switch at different times. For example, if bus *A* switches on the rising edge of the clock and bus *B* switches on the falling edge of the clock, by interleaving the bits of the two busses you can guarantee that both neighbors are constant during a switching event. This avoids the delay impact of coupling; however, you must still ensure that coupling noise does not exceed noise budgets. Figure 6.28 shows wires shielded (a) on one side, (b) on both sides, and (c) interleaved. Critical signals such as clocks or analog voltages can be shielded above and below as well.

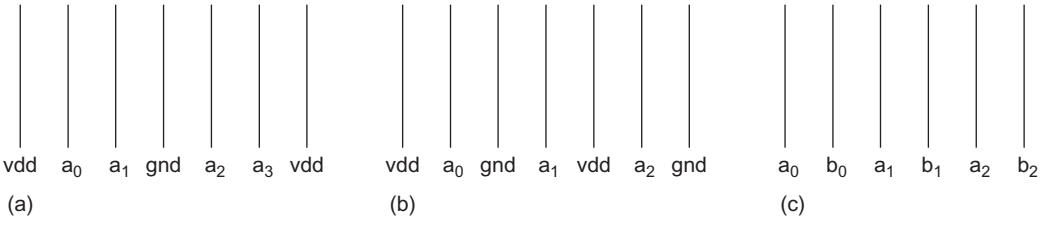


FIGURE 6.28 Wire shielding topologies

Alternatively, wires can be arranged to cancel the effects of crosstalk. Three such methods include *staggered repeaters*, *charge compensation*, and *twisted differential signaling* [Ho03b]. Each technique seeks to cause equal amounts of positive and negative crosstalk on the victim, effectively producing zero net crosstalk.

Figure 6.29(a) shows two wires with staggered repeaters. Each segment of the victim sees half of a rising aggressor segment and half of a falling aggressor segment. Although the cancellation is not perfect because of delays along the segments, staggering is a very effective approach. Figure 6.29(b) shows charge compensation in which an inverter and transistor are added between the aggressor and victim. The transistor is connected to behave as a capacitor. When the aggressor rises and couples the victim upward, the inverter falls and couples the victim downward. By choosing an appropriately sized compensation transistor, most of the noise can be canceled at the expense of the extra circuitry. Figure 6.29(c) shows twisted differential signaling in which each signal is routed differentially. The signals are swapped or *twisted* such that the victim and its complement each see equal coupling from the aggressor and its complement. This approach is expensive in wiring resources, but it effectively eliminates crosstalk. It is widely used in memory designs that are naturally differential, as explored in Section 12.2.3.3.

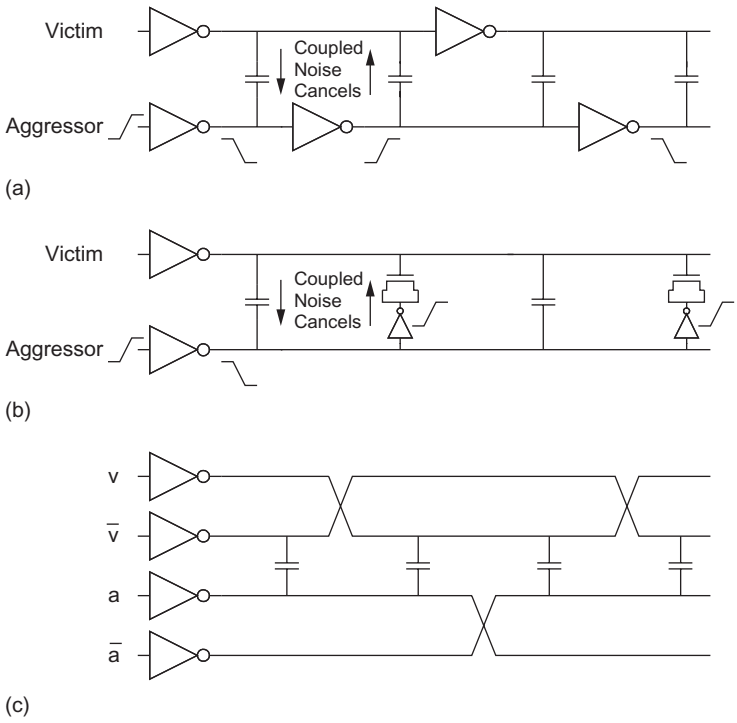


FIGURE 6.29 Crosstalk control schemes



6.4.4 Low-Swing Signaling

Driving long wires is slow because of the RC delay, and expensive in power because of the large capacitance to switch. Low-swing signaling improves performance by sensing when a wire has swung through some small V_{swing} rather than waiting for a full swing. If the driver is turned off after the output has swung sufficiently, the power can be reduced as well. However, the improvements come at the expense of more complicated driver and receiver circuits. Low-swing signaling may also require a twisted differential pair of wires to eliminate common-mode noise that could corrupt the small signal.

The power consumption for low-swing signaling depends on both the driver voltage V_{drive} and the actual voltage swing V_{swing}. Each time the wire is charged and discharged, it consumes $Q = CV_{swing}$. If the effective switching frequency of the wire is αf, the average current is

$$I_{avg} = \frac{1}{T}\int_0^T i_{drive}(t)\,dt = \alpha f C V_{swing} \tag{6.37}$$

Hence, the dynamic dissipation is

$$P_{dynamic} = I_{avg} V_{drive} = \alpha f C V_{swing} V_{drive} \tag{6.38}$$

In contrast, a rail-to-rail driver uses $V_{drive} = V_{swing} = V_{DD}$ and thus consumes power proportional to V_{DD}^2. V_{swing} must be less than or equal to V_{drive}. By making V_{swing} less than V_{drive}, we speed up the wire because we do not need to wait for a full swing. By making both voltages significantly less than V_{DD}, we can reduce the power by an order of magnitude.

Low-swing signaling involves numerous challenges. A low V_{drive} must be provided to the chip and distributed to low-swing drivers. The signal should be transmitted on differential pairs of wires that are twisted to cancel coupling from neighbors and equalized to prevent interference from the previous data transmitted. The driver must turn on long enough to produce V_{swing} at the far end of the line, then turn off to prevent unnecessary power dissipation. This generally leads to a somewhat larger swing at the near end of the line. The receiver must be clocked at the appropriate time to amplify the differential signal. Distributing a self-timed clock from driver to receiver is difficult because the distances are long, so the time to transmit a full-swing clock exceeds the time for the data to complete its small swing.

Figure 6.30 shows a synchronous low-swing signaling technique using the system clock for both driver and receiver [Ho03a]. During the first half of the cycle, the driver is OFF (high impedance) and the differential wires are equalized to the same voltage. During the second half of the cycle, the drivers turn ON. At the end of the cycle, the receiver senses the differential voltage and amplifies it to full-swing levels. Figure 6.30(a) shows the overall system architecture. Figure 6.30(b) shows the driver for one of the wires. The gates use ordinary V_{DD} while the drive transistors use V_{drive}. Because $V_{drive} < V_{DD} - V_t$, nMOS transistors are used for both the pullup and pulldown to deliver low effective resistance in their linear regime. A second driver using the complementary input drives the complementary wire. Figure 6.30(c) shows the differential wires with twisting and equalizing. The end of the wire only swings part-way, reducing power consumption. Using medium V_{drive} and small V_{swing} is faster than using a smaller V_{drive} and waiting for the wire to swing all the way. Figure 6.30(d) shows the clocked sense amplifier based on the SA-F/F that will be described further in Section 10.3.8. The sense amplifier uses pMOS input transistors because the small-swing inputs are close to GND and below the threshold of nMOS transistors. Note that the clock period must be long enough to transmit an adequate voltage swing. If the clock period increases, the circuit will actually dissipate more power because the voltage swing will increase to a maximum of V_{drive}.

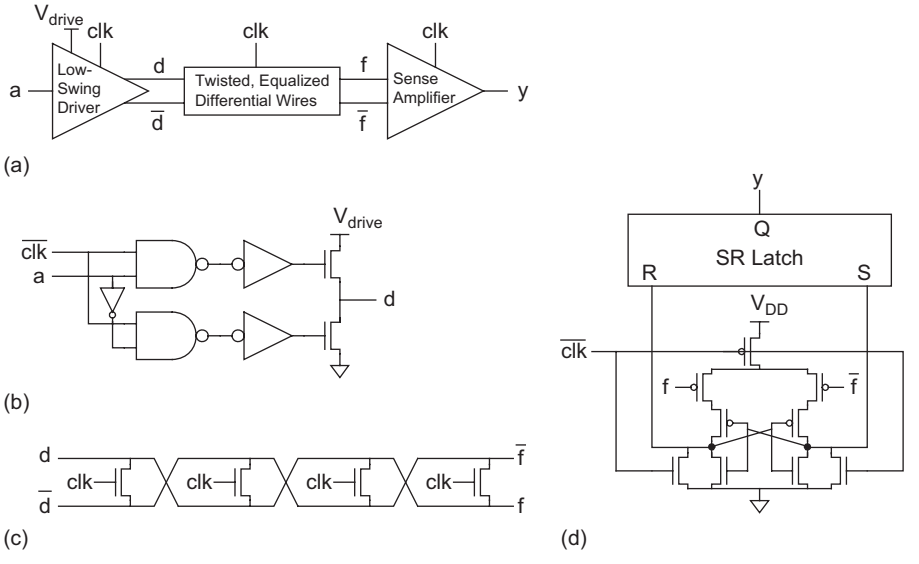


FIGURE 6.30 Low-swing signaling system

6.4.5 Regenerators

Repeaters are placed in series with wires and thus are limited to unidirectional busses. An alternative is to use *regenerators* (also called *boosters*) placed in parallel with wires at periodic intervals, as shown in Figure 6.31. When the wire is initially '0,' the regenerator senses a rising transition and accelerates it. Conversely, when the wire is initially '1,' the regenerator accelerates the falling transition. Regenerators trade off up to 20% better delay or energy for reduced noise margins.

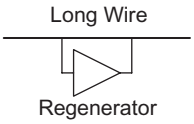


FIGURE 6.31 Regenerator

Regenerators generally use skewed gates to sense a transition. As discussed in Section 9.2.1.5, a HI-skew gate favors the rising output by using a low switching point, and a LO-skew gate does the reverse. Figure 6.32 shows a self-timed regenerator [Dobbalaere95]. When the wire begins to rise, the LO-skewed NAND gate detects the transition midway and turns on the pMOS driver to assist. The normal-skew inverters eventually detect the transition and flip node *x*, turning off the pMOS driver. When the wire begins to fall, the HI-skewed NOR gate turns on the nMOS to assist. Other regenerator designs include [Nalamalpu02, Singh08].

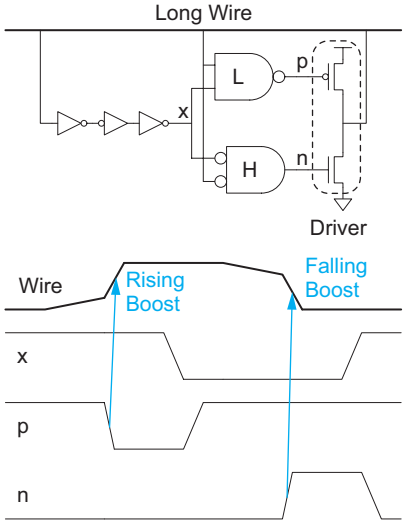


FIGURE 6.32 Regenerator

6.5 Logical Effort with Wires

Interconnect complicates the application of Logical Effort because the wires have a fixed capacitance. The branching effort at a wire with capacitance C_{wire} driving a gate load of C_{gate} is $(C_{gate} + C_{wire}) / C_{gate}$. This branching effort is not constant; it depends on the size of the gate being driven. The simple rule that circuits are fastest when all stages bear equal effort is no longer true when wire capacitance is introduced. If the wire is very short or very long, approximations are possible, but when the wire and gate loads are comparable, there is no simple method to determine the best stage effort.

Every circuit has some interconnect, but when the interconnect is short ($C_{wire} << C_{gate}$), it can be ignored. Alternatively, you can compute the average ratio of wire capacitance to parasitic diffusion capacitance and add this as extra parasitic capacitance when determining parasitic delay. For connections between nearby gates, this generally leads to a best stage effort ρ slightly greater than 4. The path should use fewer stages because each stage contributes wire capacitance. To reduce delay, the gates should be sized larger so that the wire capacitance is a smaller fraction of the whole. However, this comes at the expense of increased energy.

Conversely, when the interconnect is long ($C_{wire} >> C_{gate}$), the gate at the end can be ignored. The path can now be partitioned into two parts. The first part drives the wire while the second receives its input from the wire. The first part is designed to drive the load capacitance of the wire; the extra load of the receiver is negligible. To save energy, the final stage driving the wire should have a low logical effort and a high electrical effort; an inverter is preferred [Stan99]. The size of the receiver is chosen by practical considerations: Larger receivers may be faster, but they also cost area and power. If the wire is long enough that the RC flight time exceeds a few gate delays, it should be broken into segments driven by repeaters.

The most difficult problems occur when $C_{wire} \approx C_{gate}$. These medium-length wires introduce branching efforts that are a strong function of the size of the gates they drive. Writing a delay equation as a function of the gate sizes along the path and the wire capacitance results in an expression that can be differentiated with respect to gate sizes to com-

pute the best sizes. Alternatively, a convex optimizer can be used to minimize delay or generate an energy-delay trade-off curve.

Figure 6.33 shows three stages along a path. By writing the Elmore delay and differentiating with respect to the size of the middle stage, we find the interesting result that the delay caused by the capacitance of a stage should equal the delay caused by the resistance of the stage [Morgenshtein09]:

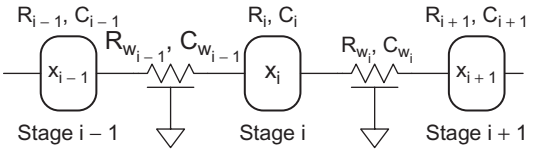


FIGURE 6.33 Path with wires

$$C_i\left(R_{i-1}+R_{w_{i-1}}\right)=R_i\left(C_{i+1}+C_{w_{i+1}}\right) \tag{6.39}$$

Example 6.11

The path in Figure 6.34 contains a medium-length wire modeled as a lumped capacitance. Write an equation for path delay in terms of x and y. How large should the x and y inverters be for shortest path delay? What is the stage effort of each stage?

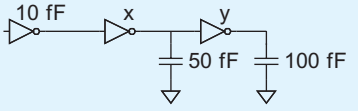


FIGURE 6.34 Path with medium-length wire

SOLUTION: From the Logical Effort delay model, we find the path delay is

$$d=\frac{x}{10}+\frac{y+50}{x}+\frac{100}{y}+P \tag{6.40}$$

Differentiating with respect to each size and setting the results to 0 allows us to solve EQ (6.41) for $x = 33$ fF and $y = 57$ fF.

$$\frac{1}{10}-\frac{y+50}{x^2}=0\Rightarrow x^2=10y+500$$
$$\frac{1}{x}-\frac{100}{y^2}=0\Rightarrow y^2=100x \tag{6.41}$$

The stage efforts are (33/10) = 3.3, (57 + 50)/33 = 3.2, and (100/57) = 1.8. Notice that the first two stage efforts are equal as usual, but the third stage effort is lower. As x already drives a large wire capacitance, y may be rather large (and will bear a small stage effort) before the incremental increase in delay of x driving y equals the incremental decreases in delay of y driving the output.

6.6 Pitfalls and Fallacies

Designing a large chip without considering the floorplan

In the mid-1990s, designers became accustomed to synthesizing a chip from HDL and "tossing the netlist over the wall" to the vendor who would place & route it and manufacture the chip. Many designers were shielded from considering the physical implementation. Now flight

times across the chip are a large portion of the cycle time in slow systems and multiple cycles in faster systems. If the chip is synthesized without a floorplan, some paths with long wires will be discovered to be too slow after layout. This requires resynthesis with new timing constraints to shorten the wires. When the new layout is completed, the long wires simply show up in different paths. The solution to this convergence problem is to make a floorplan early and microarchitect around this floorplan, including budgets for wire flight time between blocks. Algorithms termed *timing directed placement* have alleviated this problem, resulting in place & route tools that converge in one or a few iterations.

Leaving gaps in the power grid

Current always flows in loops. Current flowing along a signal wire must return in the power/ground network. The area of the loop sets the inductance of the signal. A discontinuity in the power grid can force return current to find a path far from the signal wire, greatly increasing the inductance, which increases delay and noise. Because signal inductance is usually not modeled, the delay and noise will not be discovered until after fabrication.

Summary

As feature size decreases, transistors get faster but wires do not. Interconnect delays are now very important. The delay is again estimated using the Elmore delay model based on the resistance and capacitance of the wire and its driver and load. The wire delay grows with the square of its length, so long wires are often broken into shorter segments driven by repeaters. Vast numbers of wires are required to connect all the transistors, so processes provide many layers of interconnect packed closely together. The capacitive coupling between these tightly packed wires can be a major source of noise in a system. These challenges are managed by using many metal layers of various thicknesses to provide high bandwidth for short thin wires and lower delay for longer fat wires. The microarchitecture becomes inherently linked to the floorplan because the design must allocate one or more cycles of pipeline delay for wires that cross the chip.

Exercises

6.1 Estimate the resistance per mm of a minimum pitch Cu wire for each layer in the Intel 45 nm process described in Table 6.1. Assume a 10 nm high-resistance barrier layer and negligible dishing.

6.2 Consider a 5 mm-long, 4 λ-wide metal2 wire in a 0.6 μm process. The sheet resistance is 0.08 $\Omega/\square$ and the capacitance is 0.2 fF/μm. Construct a 3-segment π-model for the wire.

6.3 A 10x unit-sized inverter drives a 2x inverter at the end of the 5 mm wire from Exercise 6.2. The gate capacitance is $C = 2$ fF/μm and the effective resistance is $R = 2.5\ \text{k}\Omega \cdot \mu\text{m}$ for nMOS transistors. Estimate the propagation delay using the Elmore delay model; neglect diffusion capacitance.

6.4 Find the best width and spacing to minimize the RC delay of a metal2 bus in the 180 nm process described in Figure 6.12 if the pitch cannot exceed 960 nm. Minimum width and spacing are 320 nm. First, assume that neither adjacent bit is switching. How does your answer change if the adjacent bits may be switching?

6.5 Derive EQ (6.27)–(6.30). Assume the initial driver and final receiver are of the same size as the repeaters so the total delay is N times the delay of a segment.

6.6 Revisit Exercise 6.5 using a pair of inverters (a noninverting buffer) instead of a single inverter. The first inverter in each pair is $W1$ times unit width. The second is a factor of k larger than the first. Derive EQ (6.33)–(6.36).

6.7 Compute the characteristic velocity (delay per mm) of a repeated metal2 wire in the 180 nm process. A unit nMOS transistor has resistance of 2.5 kΩ and capacitance of 0.7 fF, and the pMOS has twice the resistance. Use the data from Figure 6.12. Consider both minimum pitch and double-pitch (twice minimum width and spacing) wires. Assume solid metal above and below the wires and that the neighbors are not switching.

6.8 Prove EQ (6.39).

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